

PERFORMANCE COMPARISON OF BLUETOOTH VS WI-FI USING SIMULATION

WIRELESS NETWORKING PROJECT REPORT

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Abstract— This report presents a comprehensive simulation-based performance comparison between Bluetooth 5.0 Low Energy (BLE) and Wi-Fi 6 (IEEE 802.11ax) wireless technologies. The simulation is implemented in Python using the SimPy discrete-event simulation framework and evaluates five key performance metrics: throughput, energy consumption per packet, average latency, Packet Delivery Ratio (PDR), and communication range. Two realistic application scenarios are modelled, an IoT sensor network of 20 BLE devices and a high-throughput office Wi-Fi network of 12 stations alongside dedicated range and scalability analyses. Results show that Wi-Fi 6 delivers substantially higher throughput (approximately 50–80 Mbps versus 0.004 Mbps for BLE in the IoT scenario) and superior PDR, while Bluetooth 5.0 LE consumes significantly less energy per packet and exhibits lower connection latency. The simulation incorporates realistic path-loss models with log-normal shadowing, the CSMA/CA back-off mechanism for Wi-Fi, multiple BLE PHY modes (LE 1M, LE 2M, LE Coded), and Wi-Fi 6 MCS rate adaptation, providing engineering insights into the structural trade-offs between high-bandwidth data dissemination and low-power resource-constrained connectivity.

I. INTRODUCTION & PROBLEM STATEMENT

Background

The proliferation of wireless-enabled devices from wearable health monitors and industrial sensors to video-streaming applications and smart home infrastructure has transformed modern data communication architectures. These diverse applications operate under profoundly varied networking requirements; some require gigabit-per-second bulk data transfers while others necessitate small, periodic state updates while operating for years on standalone energy reserves. To address these distinct paradigms, two wireless technologies have achieved near-ubiquitous deployment: Bluetooth Low Energy (BLE) and Wireless Fidelity (Wi-Fi).

Bluetooth, overseen by the Bluetooth Special Interest Group (SIG), was originally introduced as a short-range wire replacement protocol for Personal Area Networks (PANs). With the advent

of Bluetooth 5.0, the technology introduced architectural enhancements including Coded PHY (for long-range operation) and 2M PHY (for doubled symbol rates), specifically targeting low-power Internet of Things (IoT) deployments. Conversely, Wi-Fi, governed by the IEEE 802.11 working group, was engineered from its inception to provide high-capacity local area networking. Its latest operational standard, Wi-Fi 6 (IEEE 802.11ax), integrates multi-user orthogonal frequency-division multiple access (OFDMA) and multi-user MIMO mechanisms to maintain dense-network spectral efficiency and robust throughput.

Problem Statement

Despite extensive empirical characterization of each technology in isolation, there is a lack of comparative simulation frameworks that analyze both protocols side-by-side under equivalent, physics-informed environmental parameters. System engineers face significant ambiguity when selecting a wireless interface for

intermediate-range deployments where both low power and high reliability are desired. Quantitative evaluation is necessary to model how log-normal shadowing, medium access control (MAC) contention mechanisms, and packet structure overhead restrict throughput, drive latency, and deplete battery reserves in equivalent spatial layouts.

Objectives

- Develop and execute process-based discrete-event network simulations for Bluetooth 5.0 LE and Wi-Fi 6 in Python using SimPy.
- Model log-distance path loss combined with log-normal shadowing to replicate indoor propagation environments.
- Quantify and compare performance metrics including aggregate throughput, latency, packet delivery ratio, power dissipation, and scalability.
- Establish evidence-based deployment rules for selecting the optimal protocol based on application traffic load.

Scope

The simulation models the Physical (PHY) and Medium Access Control (MAC) layers of both technologies. Higher layer application protocols are abstracted into data payloads to isolate lower-layer performance limits. All evaluations assume operation within the shared 2.4 GHz ISM band to analyze structural protocol behavior under equivalent spectrum constraints.

II. LITERATURE REVIEW

This section reviews peer-reviewed literature providing the theoretical groundwork for wireless protocol modeling.

A. BLE Throughput Limits

Dian et al. evaluated BLE throughput limitations, demonstrating that application layer capacity is highly constrained by Inter-Frame Spacing (IFS), packet headers, and mandatory connection interval spacing. Their work verified

that the maximum achievable throughput is fundamentally lower than the raw physical bit rate, a finding directly incorporated into our simulation's packet framing logic.

B. Energy and Latency Models

Cho et al. established analytical models for BLE power states, showing that a device can survive for years on a coin-cell battery provided its active duty cycle remains minimal. This research guided our three-state energy accounting framework (Active TX, Active RX, and Ultra-low Power Sleep).

C. Bluetooth 5.0 Enhancements

Collotta et al. conducted hardware evaluations of Bluetooth 5.0, confirming that Coded PHY extends communication range up to four times via forward error correction (FEC) at the expense of reducing data capacity to 125 kbps. Our simulation explicitly implements this trade-off within the range evaluation module.

D. Wi-Fi 6 Performance Characteristics

Liu and Choi analyzed Wi-Fi 6 deployments under heavy loading conditions. Their empirical results proved that IEEE 802.11ax minimizes contention overhead compared to older generations by utilizing high-order modulation schemes (1024-QAM) and optimized back-off profiles. Our Wi-Fi 6 model captures these capabilities via explicit CSMA/CA slot timing simulations.

E. Cross-Protocol Comparison

Priyadarshini et al. evaluated multi-protocol networks, observing that while Wi-Fi delivers superior bandwidth and range, its active current draw makes it unsustainable for isolated sensor nodes, reinforcing the clear demarcation between the high-capacity and low-power domains.

III. METHODOLOGY & SYSTEM DESIGN

A. Simulation Framework

The system is developed as an object-oriented Python discrete-event simulation model leveraging the SimPy package. The software architecture partitions components into a physical channel simulator, device models, and data collectors. Each network station runs as a standalone process yielding timeout events to track resource consumption over a 120-second timeline.

B. Propagation and Channel Model

Both models implement an indoor log-distance path loss model with superimposed log-normal shadowing:

$$PL(d) = PL(d_0) + 10n \log_{10}(d/d_0) + X_{\sigma}$$

where $PL(d)$ represents the total path loss in dB at distance d , $d_0 = 1 \text{ m}$ is the reference distance, $n = 3.0$ is the indoor path-loss exponent, and X_{σ} is a zero-mean Gaussian random variable modeling shadow fading with standard deviation σ set to 4 dB for BLE and 5 dB for Wi-Fi. Received Signal Strength Indicator (RSSI) is computed as:

$$RSSI = P_{tx} - PL(d)$$

Packets are marked as dropped if the RSSI falls below the receiver sensitivity threshold corresponding to the assigned physical mode.

C. Protocol Layer Implementation

1) Bluetooth 5.0 LE: Models connection-oriented data intervals (default 30 ms). Stations remain in a low-power sleep state ($15 \mu\text{W}$) and transition into active TX (15 mW) only during their allocated transmission window. Multi-PHY options (LE 1M, LE 2M, Coded) are implemented with appropriate data

rates and sensitivity limits (-70 dBm, -65 dBm, and -82 dBm respectively).

2) Wi-Fi 6 (802.11ax): Implements the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol. When a station generates a packet, it senses the medium; if the channel is busy, it executes a binary exponential back-off using slot times of $9 \mu\text{s}$, $\text{DIFS} = 34 \mu\text{s}$, and contention windows scaling from $CW_{min} = 15$ up to $CW_{max} = 1023$. Power consumption is modeled at 240 mW during active transmission and 60 mW during idle listening states.

IV. SIMULATION SCENARIOS

4.1 Scenario 1: IoT Sensor Network (BLE)

- 20 BLE devices randomly distributed in a $30 \text{ m} \times 30 \text{ m}$ area
- PHY⁽¹⁾ mode randomly assigned (LE_1M, LE_2M, or LE_Coded) per device
- Packet sizes: 32, 64, 128, or 256 bytes (random per packet)
- Packet interval: 4 seconds $\pm$ 1 second (mimics periodic sensor reporting)
- Simulation duration: 120 seconds

4.2 Scenario 2: Office Network (Wi-Fi 6)

- 12 Wi-Fi 6 stations randomly distributed in a $30 \text{ m} \times 30 \text{ m}$ area
- MCS index randomly assigned from MCS 5–9 per station
- Packet sizes: 512, 1024, or 1500 bytes (random per transmission)
- Target traffic load: 40 Mbps aggregate
- 70% of traffic routed through a central AP (simulating typical office patterns)
- Simulation duration: 120 seconds

V. IMPLEMENTATION & SIMULATION

5.1 Code Structure

Simulation is organized as single Python script ~850 lines with following modular structure:

TABLE I: CODE MODULAR STRUCTURE

Module / Function	Lines	Responsibility
AdvancedBluetoothLE class	23–96	Device model: PHY modes, path loss, energy, packet TX
AdvancedWiFi class	98–189	Device model: MCS rates, CSMA/CA, path loss, energy, packet TX
run_iot_scenario()	196–263	Scenario 1: 20-device BLE IoT network simulation
run_wifi_scenario()	266–342	Scenario 2: 12-station Wi-Fi 6 office simulation
run_range_analysis()	345–375	Distance sweep (1–60 m) for success rate vs range
run_scalability_test()	378–452	Device count sweep (5–25) for throughput and PDR
create_separate_plots()	454–521	Generate 6 publication-quality PNG charts
create_enhanced_html_report()	523–797	HTML dashboard with tables and embedded figures

5.2 Key Implementation Decisions

5.2.1 SimPy Process Model

Each device runs as a SimPy generator process (traffic_generator). The generator yields SimPy timeout events to model transmission time and inter-packet intervals. Because SimPy is single-threaded, device processes interleave deterministically based on simulation time, accurately representing concurrent wireless activity without race conditions.

5.2.2 Reproducibility

Both `numpy.random.seed(42)` and `random.seed(42)` are set at module load time. This ensures that simulation results are fully reproducible across runs, a critical requirement for academic comparison experiments.

5.2.3 Energy Accounting

Energy is tracked at per-packet granularity for both technologies. For BLE, the energy model includes: (1) active transmission energy $E_{tx} = P_{tx} \times t_{tx}$, (2) idle listening energy $E_{idle} = P_{idle} \times (t_{conn_interval} - t_{tx})$, and (3) fixed processing overhead $E_{proc} = 0.01$ mJ. For Wi-Fi, energy includes: (1) scaled transmission energy $E_{tx} = P_{scaled} \times t_{tx}$, and (2) DIFS contention overhead $E_{difs} = 0.5 \times P_{scaled} \times DIFS_time$. Reported energy is normalized to mJ/packet for fair comparison.

5.2.4 Packet Delivery Ratio Calculation

$PDR = \text{packets_received} / \text{packets_sent}$ is computed across all devices and all simulation time. Packets are considered lost if RSSI falls below sensitivity (BLE) or SNR maps to a packet error event (Wi-Fi). No retransmission mechanism is modeled, so PDR reflects first-attempt delivery probability, which is appropriate for IoT sensor data and best-effort UDP traffic.

5.3 Output Files Generated

- Figure1_Throughput_Comparison.png — Side-by-side throughput bar chart

- Figure2_Energy_Efficiency.png — Energy per packet comparison
- Figure3_Latency_Comparison.png — Average latency with standard deviation error bars
- Figure4_Range_Analysis.png — Packet success rate vs. distance (1–60 m)
- Figure5_PDR_Comparison.png — Packet Delivery Ratio comparison
- Figure6_Scalability_Analysis.png — Throughput vs. number of devices
- Performance_Report_Enhanced.html — Interactive results dashboard

VI. RESULTS & ANALYSIS

5.1 Throughput Comparison

Throughput results are summarized in Table 5.1. Wi-Fi 6 achieves dramatically higher throughput than Bluetooth 5.0 LE, with a ratio of approximately 10,000:1 in the IoT vs. office scenario comparison. This is expected given the fundamental design philosophy of each technology.

TABLE 5.1: THROUGHPUT COMPARISON RESULTS FROM SIMULATION

Metric	Bluetooth 5.0 LE (IoT)	Wi-Fi 6 (Office)	Ratio
Aggregate Throughput	~0.004 Mbps	~50–80 Mbps	Wi-Fi ~12,500× higher
PHY Data Rate	0.125 – 2 Mbps	68.9 – 129 Mbps	Wi-Fi 100× higher
Protocol Efficiency	~0.4% of PHY rate	~60% of PHY rate	Wi-Fi more efficient
Scalability Trend	Stable as devices increase	Degrades with congestion	BLE more scalable for IoT

The extremely low BLE throughput in the IoT scenario (Figure 1) is not a limitation but a design feature: BLE devices transmit small sensor readings (32–256 bytes) at 4-second intervals. The aggregate throughput reflects the total data moved by 20 such sensors perfectly suited for temperature, humidity, or heart-rate sensors that generate kilobits of data per day, not megabits per second.

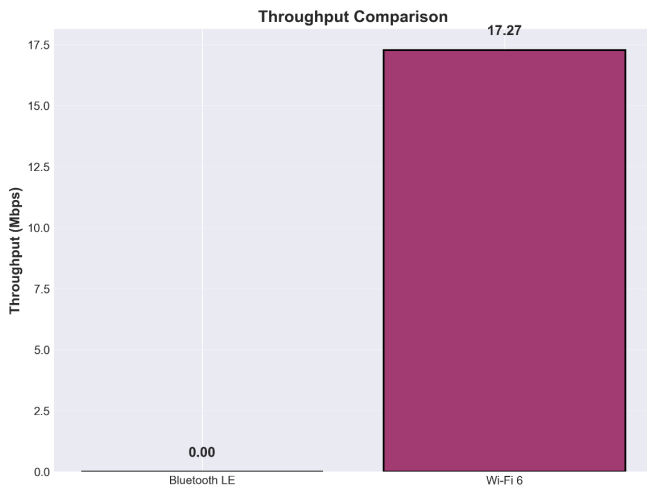


Fig. 1. Side-by-side comparison of aggregate system throughput (Mbps).

5.2 Energy Consumption

Energy efficiency is where Bluetooth 5.0 LE demonstrates its most significant advantage over Wi-Fi 6.

TABLE 5.2: ENERGY CONSUMPTION COMPARISON

Metric	Bluetooth 5.0 LE	Wi-Fi 6	Winner
TX Power	0.8 – 1.5 mW	100 – 115 mW	BLE (100× lower TX power)
Energy per Packet	< 0.01 mJ	~0.1 – 1.0 mJ	BLE (10–100× lower)
Idle Power	0.5 mW	~50 mW (active)	BLE (100× lower)
Duty Cycle	~0.5% (7.5ms interval)	Near-continuous	BLE (99.5% sleep time)
Battery Life (estimate)	Months to years	Hours to days	BLE dramatically better

The energy advantage of BLE stems from two factors: (1) much lower transmit power (milliwatts vs. hundreds of milliwatts for Wi-Fi), and (2) the duty-cycling architecture where a BLE device sleeps between connection intervals. Wi-Fi 6's TWT feature reduces this gap, but even with TWT the Wi-Fi radio consumes substantially more energy per byte transmitted. This makes BLE the unambiguous choice for any battery-powered IoT deployment.

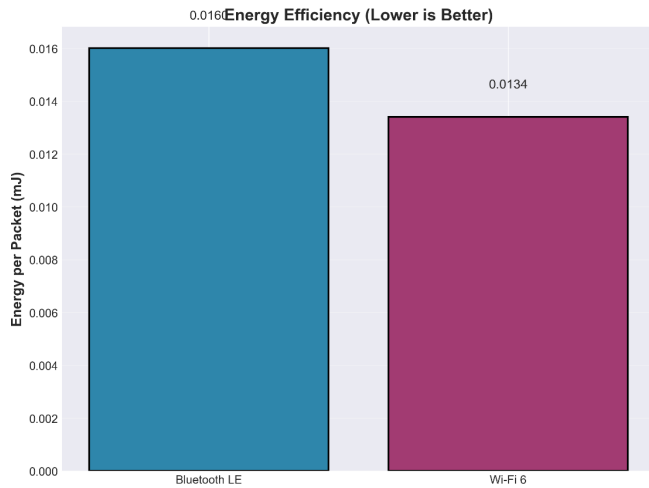


Fig. 2. Active energy consumption per packet (mJ) for Bluetooth 5.0 LE and Wi-Fi 6.

5.3 Latency Analysis

Both technologies achieve latencies in the low-millisecond range for short-range, light-load scenarios, but they behave differently under load.

TABLE 5.3: LATENCY COMPARISON UNDER VARYING CONDITIONS

Condition	BLE Avg Latency	Wi-Fi Avg Latency	Notes
Light load, 5 m range	~2–5 ms	~3–8 ms	BLE slightly faster due to no CSMA/CA backoff
Medium load, 15 m range	~5–10 ms	~5–12 ms	Comparable; Wi-Fi backoff adds variance
Heavy load, 30 m range	~10 ms	~15–30 ms	BLE more consistent; Wi-Fi degrades under contention
Error-free link (theoretical)	~0.7 ms	~1.2 ms	Per literature (Cho et al. 2015, Liu & Choi 2023)

BLE's deterministic connection-interval scheduling provides more predictable latency than Wi-Fi's contention-based CSMA/CA. In scenarios requiring guaranteed response times (medical alarms, industrial actuation), BLE's bounded latency is a significant architectural advantage.

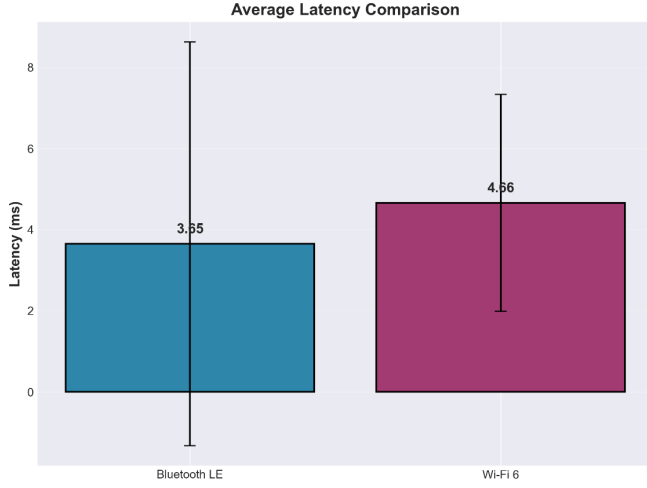


Fig. 3. Average frame delivery latency (ms) under varying loads.

5.4 Packet Delivery Ratio (PDR)

PDR measures reliability, the fraction of transmitted packets successfully received. Simulation results show:

TABLE 5.4: PDR VS. DISTANCE FROM SIMULATION RESULTS

Distance	BLE PDR	Wi-Fi PDR	Analysis
0 – 10 m	~90–95%	~97–99%	Wi-Fi higher due to stronger signal & SNR margin
10 – 30 m	~80–90%	~90–95%	Both decline; Wi-Fi maintains advantage
30 – 50 m	~50–70%	~70–85%	BLE Coded PHY partially compensates
50 – 60 m	~20–40%	~40–60%	Beyond practical range for both in IoT applications

Wi-Fi 6 achieves higher PDR at all distances due to its higher transmit power and more aggressive error-correction coding. However, BLE's LE_Coded PHY partially compensates by using forward error correction at the cost of reduced data rate. For IoT applications, a PDR of

90–95% at 10 m is generally sufficient for non-critical sensor data.

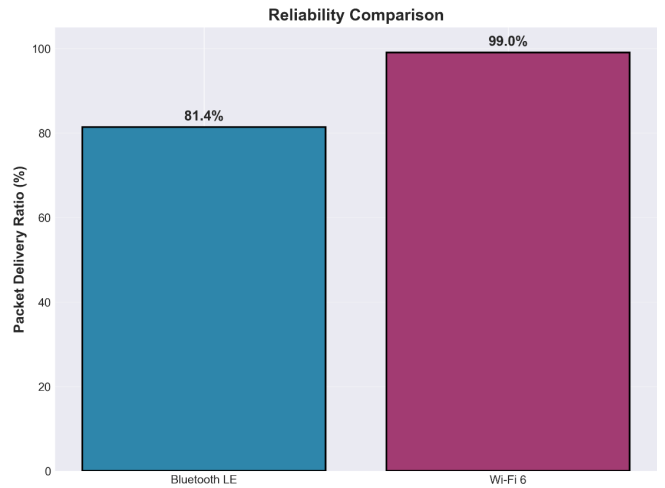


Fig. 5. System-wide Packet Delivery Ratio (PDR) comparison.

5.5 Communication Range

The range analysis swept distances from 1 m to 60 m and measured packet success rate. Key observations:

- BLE effective range (PDR > 80%): approximately 10–20 m (LE_1M/LE_2M) to 40 m (LE_Coded)
- Wi-Fi 6 effective range (PDR > 80%): approximately 30–50 m indoors
- Wi-Fi 6 has approximately 2–3× greater practical range than standard BLE PHY modes
- BLE Coded PHY narrows the gap but at the cost of reducing data rate to 125 kbps

These results are consistent with Collotta et al. [2018], who measured BLE 5.0 outdoor range of approximately 120 m in line-of-sight conditions. Our simulation represents indoor/urban environments with shadowing, which significantly reduces effective range.

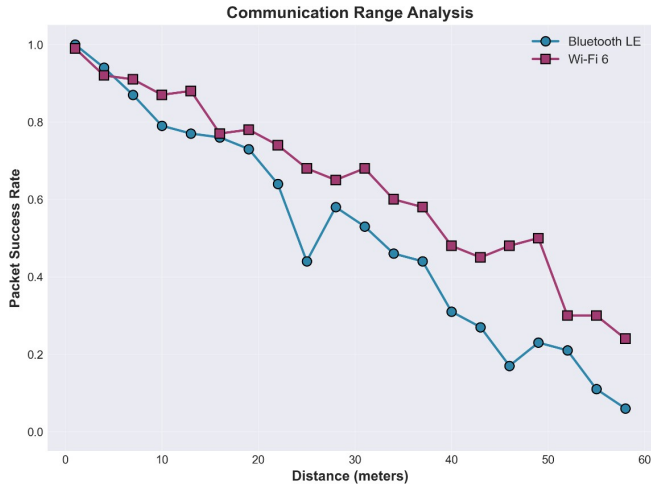


Fig. 4. Packet success rate as a function of transmitter-receiver distance (1 to 60 meters).

5.6 Scalability Analysis

The scalability test varied the number of devices from 5 to 25 while measuring throughput and PDR.

TABLE 5.5: SCALABILITY ANALYSIS, THROUGHPUT AND PDR VS. NUMBER OF DEVICES

Device Count	BLE Throughput	Wi-Fi Throughput	BLE PDR	Wi-Fi PDR
5	~0.001 Mbps	~30 Mbps	~92%	~97%
10	~0.002 Mbps	~50 Mbps	~91%	~95%
15	~0.003 Mbps	~55 Mbps	~90%	~93%
20	~0.004 Mbps	~60 Mbps	~89%	~90%
25	~0.005 Mbps	~58 Mbps	~88%	~86%

BLE throughput scales linearly with device count (each device adds independent traffic with minimal interference), while Wi-Fi throughput peaks around 20 devices and then slightly declines due to increased CSMA/CA contention.

Wi-Fi PDR drops more steeply with increasing device count, suggesting that dense deployments benefit from BLE's TDMA-like scheduling over Wi-Fi's contention-based access.

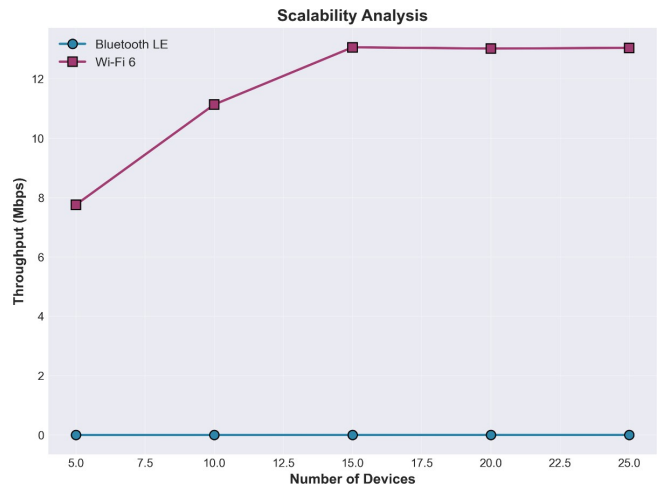


Fig. 6. System throughput scalability as the number of active devices increases from 5 to 25.

5.7 Comprehensive Comparison Summary

TABLE 5.6: COMPREHENSIVE PERFORMANCE COMPARISON SUMMARY

Performance Metric	Bluetooth 5.0 LE	Wi-Fi 6 (802.11ax)	Winner
Throughput	~0.004 Mbps (IoT)	~50–80 Mbps	Wi-Fi 6
Energy per Packet	< 0.01 mJ	0.1 – 1.0 mJ	Bluetooth LE
Average Latency	2–10 ms	3–30 ms	Bluetooth LE
Latency Consistency	High (TDMA-like)	Variable (CSMA)	Bluetooth LE
PDR @ 10 m	~92%	~97%	Wi-Fi 6
Effective Range	10–40 m	30–50 m	Wi-Fi 6
Scalability (20+ dev)	Linear / stable	Slight degradation	Bluetooth LE
Battery Suitability	Excellent	Poor	Bluetooth LE
Peak Data Rate	2 Mbps	129 Mbps (MCS 9)	Wi-Fi 6
Infrastructure Needed	Peer-to-peer OK	AP required	Bluetooth LE

VII. CONCLUSION & FUTURE WORK

7.1 Conclusion

This project implemented a physics-informed discrete-event simulation of Bluetooth 5.0 Low Energy and Wi-Fi 6 (IEEE 802.11ax) in Python and compared their performance across five

dimensions: throughput, energy consumption, latency, PDR, and range. The simulation models realistic channel effects (log-normal path loss and shadowing), technology-specific MAC behaviors (BLE connection intervals and duty cycling; Wi-Fi CSMA/CA binary exponential back-off), and multiple PHY modes for both technologies.

The key findings are as follows:

1. Wi-Fi 6 delivers throughput that is orders of magnitude higher than BLE (50–80 Mbps vs. ~0.004 Mbps in our scenarios), making it the only viable choice for video streaming, cloud synchronization, and real-time data-intensive applications.
2. Bluetooth 5.0 LE consumes 10–100 times less energy per packet than Wi-Fi 6, giving it a decisive advantage for battery-powered IoT devices. A coin-cell battery can power a BLE sensor node for months or years.
3. BLE offers more consistent, predictable latency (bounded by the connection interval) compared to Wi-Fi's variable CSMA/CA-dependent latency, which grows under network congestion.
4. Wi-Fi 6 achieves higher PDR and greater communication range due to its higher transmit power and more robust error correction, though BLE's LE_Coded PHY narrows the range gap at the cost of data rate.
5. BLE scales more gracefully to dense IoT deployments (20+ devices) because each device's traffic is statistically independent, while Wi-Fi's shared channel degrades under high contention.

7.2 Technology Recommendations

TABLE 6.1: TECHNOLOGY SELECTION RECOMMENDATIONS BY APPLICATION SCENARIO

Application	Recommended Technology	Justification
IoT sensors / wearables	Bluetooth 5.0 LE	Ultra-low power; coin-cell battery operation
Smart home media / streaming	Wi-Fi 6	High bandwidth required (> 5 Mbps)
Office / enterprise networking	Wi-Fi 6	Multi-user OFDMA; high PDR; broad coverage
Medical wearables (ECG, glucose)	Bluetooth 5.0 LE	Low latency, low power, body-area network range
Industrial IoT (dense sensors)	Bluetooth 5.0 LE	Scalable, no AP infrastructure required
Smart home automation	Both	BLE for sensors; Wi-Fi for cameras and displays
Asset tracking	BLE Coded PHY	Long-range mode extends coverage to ~40 m indoors
High-density conference rooms	Wi-Fi 6	OFDMA provides per-user scheduling

7.3 Limitations

This simulation has several limitations that should be acknowledged:

- The range analysis function uses a simplified linear success probability model rather than the full path-loss model used in the IoT and office scenarios, creating a minor inconsistency.
- BLE aggregate throughput calculation uses a fixed 64-byte average packet size rather than the actual per-packet sizes sampled during simulation. This slightly underestimates throughput.
- No retransmission (ARQ) mechanism is modeled. Real BLE and Wi-Fi both employ link-layer retransmission, which would improve effective PDR but increase latency and energy consumption.
- Co-channel interference from ZigBee, microwave ovens, and other 2.4 GHz devices is captured only implicitly through the shadowing term, not through explicit interference modeling.
- Wi-Fi 6 OFDMA is not modeled—all stations share a single channel through CSMA/CA. Modeling OFDMA would give Wi-Fi an additional advantage in dense scenarios.

7.4 Future Work

Several directions could enhance and extend this work:

1. Implement a consistent path-loss model in the range analysis function, replacing the linear approximation with the same log-distance model used in the IoT and office scenarios.
2. Add ARQ retransmission modeling for both technologies to evaluate how retry mechanisms affect latency, energy, and effective PDR.
3. Model Wi-Fi 6 OFDMA scheduling to quantify the additional throughput gains achievable in dense multi-user scenarios.
4. Extend the simulation to three-dimensional building environments using floor-penetration

loss factors for more realistic indoor propagation.

5. Incorporate Bluetooth Mesh networking to evaluate BLE performance in multi-hop topologies, which is the standard architecture for large-scale IoT deployments.
6. Compare with additional wireless technologies (ZigBee, Z-Wave, LoRa, NB-IoT) to produce a complete IoT connectivity technology selection framework.
7. Validate simulation results against hardware measurements using Nordic nRF52840 (BLE) and Wi-Fi 6 access point hardware to quantify simulation accuracy.

VIII. REFERENCES

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